

- 6 A tennis ball is thrown horizontally in air from the top of a tall building. The effect of air resistance is not negligible.

Which of the following correctly describes the subsequent change in the horizontal and vertical components of the ball's velocity?

	horizontal component of velocity	vertical component of velocity
A	constant	constant
B	constant	increases at a constant rate
C	decreases to zero	increases at a constant rate
D	decreases to zero	increases to a maximum value

Since the building is tall, the time of flight is long enough for the vertical component of velocity to reach a maximum value.

Answer: D

- 7 A rock of mass $2m$ in deep space, initially travelling at velocity u , explodes into two parts of equal